

Abstract Linear Algebra

Prof. Lanier

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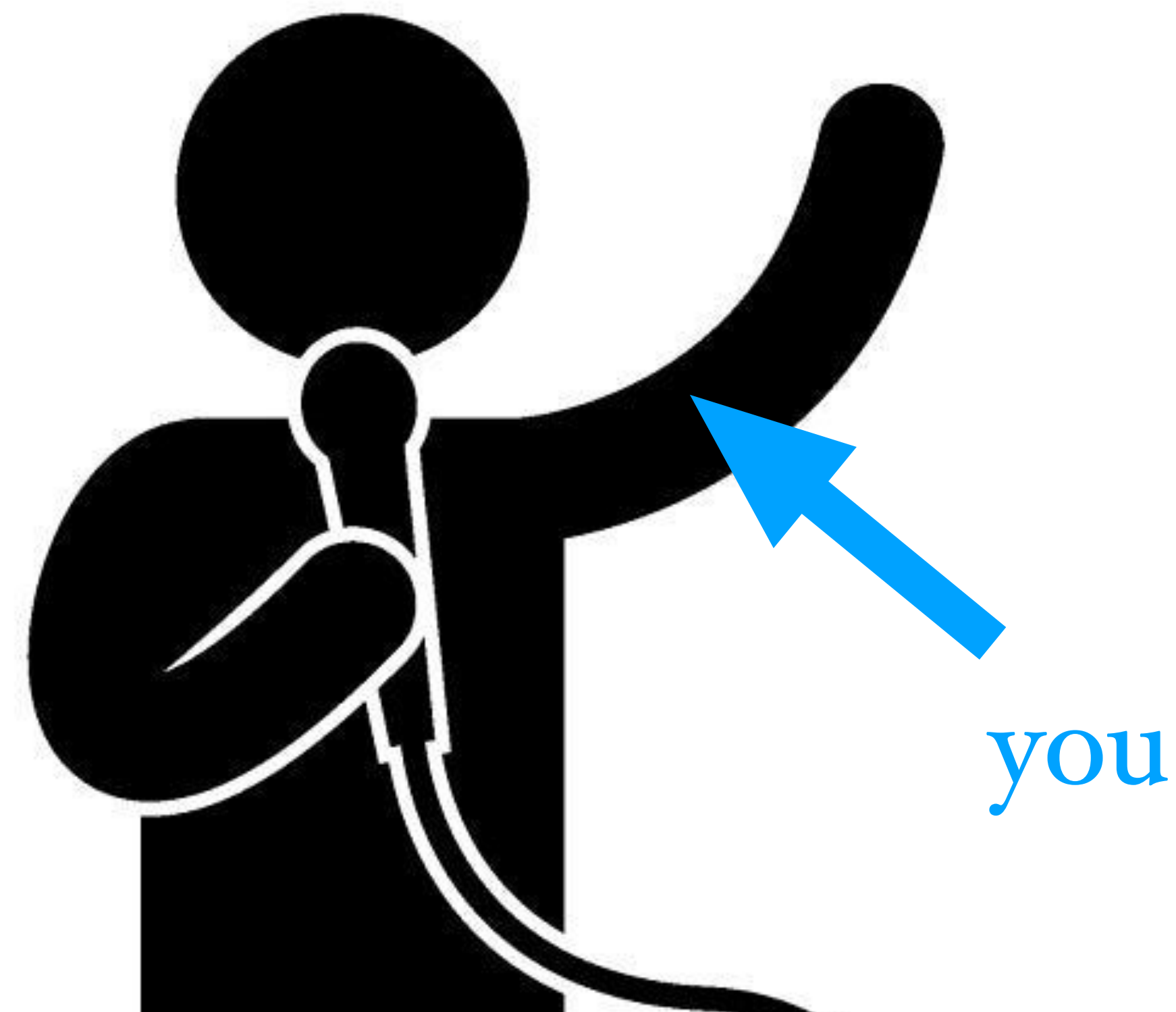
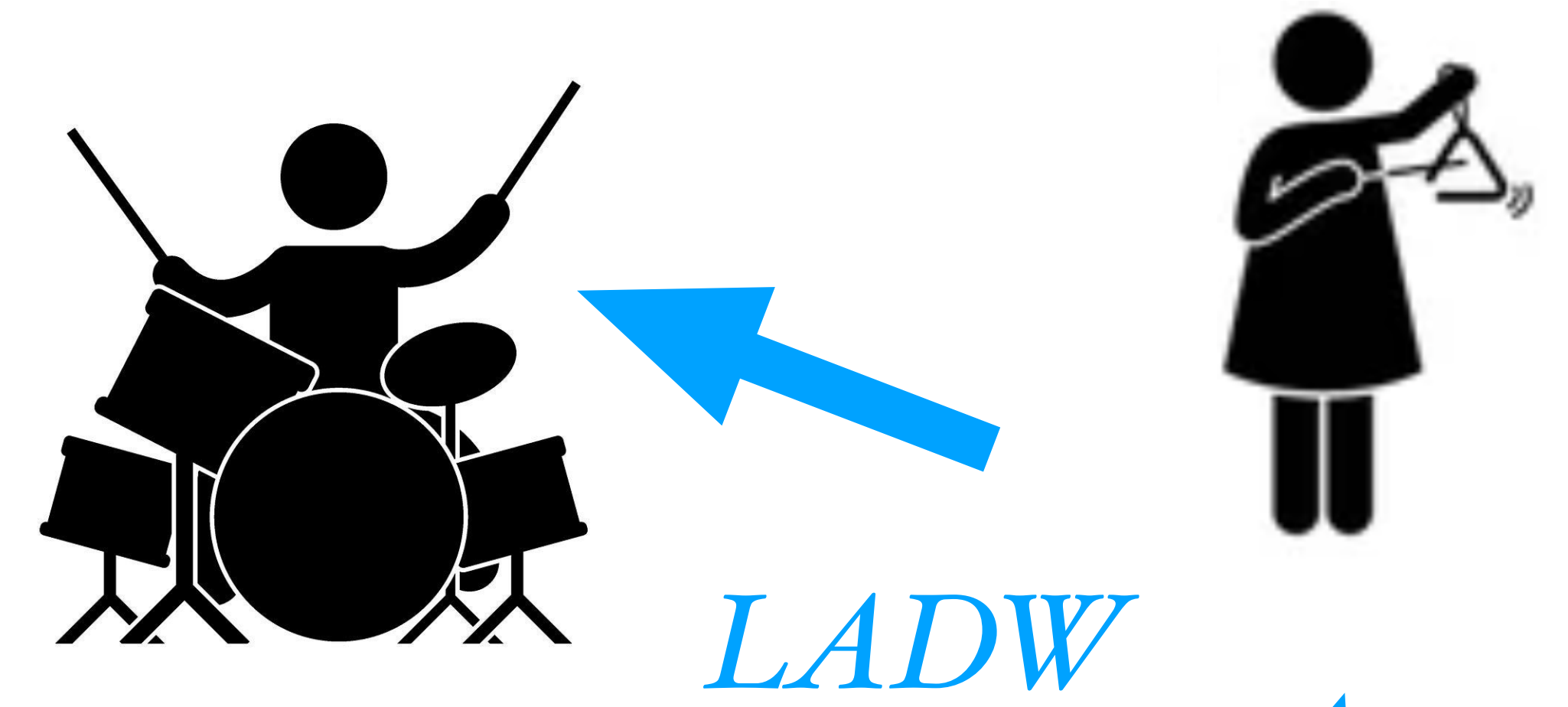
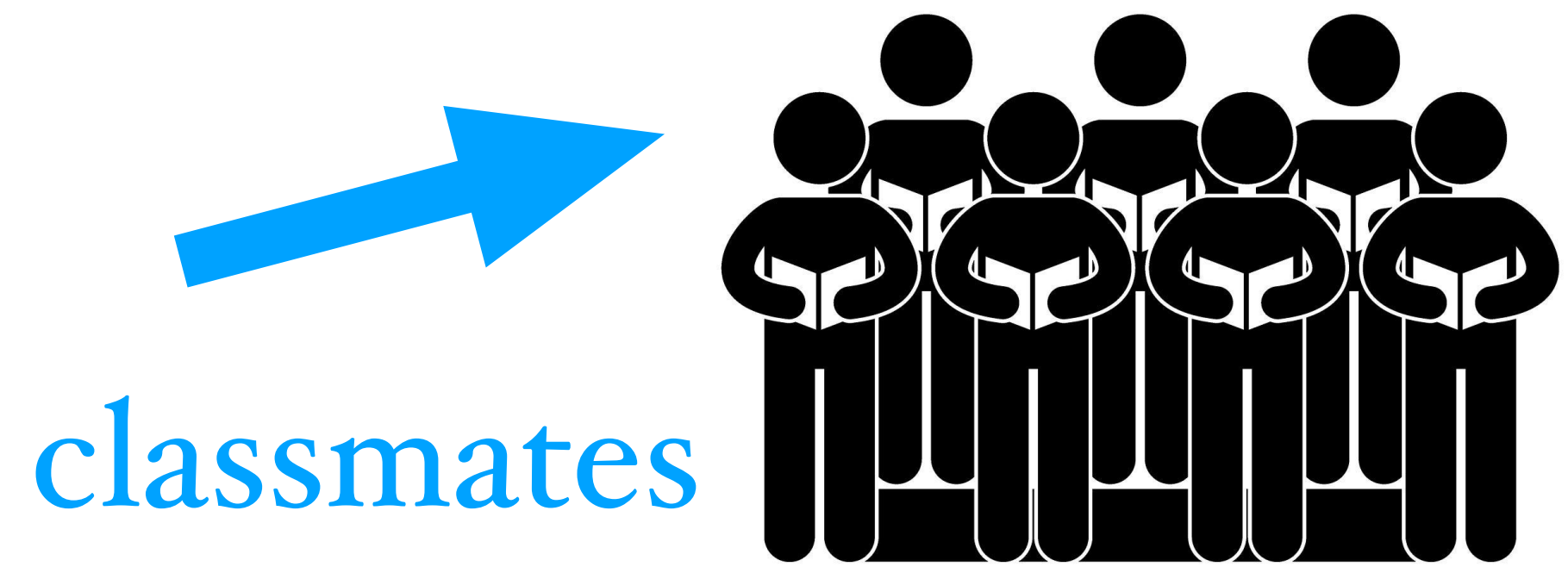
Pick up a handout. Try those warm-up problems.

Next up:

- Last requizzing session today, 3-5pm in **Eckhart 206**.
- Flex (or flex plan) due by end-of-day today.
- Final Exam review sessions: Saturday, 3:00-4:30pm, **Kent 120** (packet)
Sunday, 3:00-4:30pm, **Kent 120** (2021 final exam)
Answer keys for each are now in Files under Final.
- End-of-quarter showcase: Monday, 3:00-4:30pm, **Eckhart 202**
Sign up to present by end-of-day today.
- Final Exam: Both exams will be held in our classroom, Ryerson 276.
Section 45 (11:30am section) - Thursday, May 25, 5:30-7:30pm
Section 55 (12:30pm section) - Wednesday, May 24, 7:30-9:30am
- Please fill out the end-of-course survey and the UChicago course evaluation.

Goals of the course:

- To help you to learn the foundational ideas, skills, and results of linear algebra.
- To prepare you for upper-level math coursework, including the algebra and analysis sequences.
- To help you to grow as a mathematician.
- To make you feel both confident and stretched, during the course and once it is over.





Jordan canonical form

$$\begin{pmatrix} 8 & -20 & 14 & -118 \\ 5 & -18 & 10 & -91 \\ -35 & 147 & -60 & 558 \\ -5 & 21 & -9 & 87 \end{pmatrix}$$

arbitrary complex
operator

Jordan canonical form

$$\begin{pmatrix} 8 & -20 & 14 & -118 \\ 5 & -18 & 10 & -91 \\ -35 & 147 & -60 & 558 \\ -5 & 21 & -9 & 87 \end{pmatrix} \sim \begin{pmatrix} 3 & 1 & 3 & -25 \\ 0 & 3 & 1 & -12 \\ 0 & 0 & 3 & 35 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

arbitrary complex
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guaranteed by Schur
representation

(upper triangular)

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$$\sim \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

guaranteed by
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(block diagonal
with Jordan blocks)

Block diagonal matrices

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 0 & 5 & 6 \\ 0 & 0 & 7 & 8 \end{pmatrix} \qquad \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & \cos \psi & -\sin \psi \\ 0 & 0 & \sin \psi & \cos \psi \end{pmatrix}$$
$$\begin{pmatrix} 1 & 2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 6 & 7 & 0 \\ 0 & 0 & 8 & 9 & 10 & 0 \\ 0 & 0 & 11 & 12 & 13 & 0 \\ 0 & 0 & 0 & 0 & 0 & 14 \end{pmatrix}$$

Jordan blocks

(λ)

$$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$$

$$\begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$\begin{pmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$$

Jordan blocks

$$(\lambda) \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \quad \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} \quad \begin{pmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$$

Jordan decomposition theorem (Thm. 9.5.1): Given an operator T there exist a basis (Jordan canonical basis) such that the matrix A of T in this basis has a block diagonal form with blocks of form

$$\begin{pmatrix} \lambda & 1 & & & 0 \\ & \lambda & 1 & & \\ & & \lambda & \ddots & \\ & & & \ddots & 1 \\ 0 & & & & \lambda \end{pmatrix}$$

where λ is an eigenvalue of A . (Here we assume that the block of size 1 is just λ .) The matrix A is unique up to permutation of the blocks.

Jordan blocks

$$(\lambda) \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \quad \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} \quad \begin{pmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$$

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“as close to diagonal as possible”

$$\begin{pmatrix}
 \text{Jordan} & 1 & 0 & 0 \\
 0 & \text{Jordan} & 1 & 0 \\
 0 & 0 & \text{Jordan} & 1 \\
 0 & 0 & 0 & \text{Jordan}
 \end{pmatrix}$$

block diagonal matrix $\implies$ invariant subspaces (with a basis of subspaces)

basis of invariant subspaces $\implies$ similar to a block diagonal matrix
with blocks for the invariant subspaces

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A subspace E is A -invariant if $AE \subseteq E$.

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The kernel (a subspace) of $(A - \lambda I)^k$ for large k is the generalized eigenspace for λ .

Vectors in a generalized eigenspace for λ are called generalized eigenvectors.

$$\begin{pmatrix} 3 & 1 & 3 & -25 \\ 0 & 3 & 1 & -12 \\ 0 & 0 & 3 & 35 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 & 3 & -25 \\ 0 & 3 & 1 & -12 \\ 0 & 0 & 3 & 35 \\ 0 & 0 & 0 & 8 \end{pmatrix} \sim \begin{pmatrix} 3 & 1 & 3 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

change basis
so that an 8-eigenvector
is in the basis

$$\begin{pmatrix} 3 & 1 & 3 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix} \rightarrow \text{Null} \begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{has nullity 1, rank 2}$$

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Proof sketch: Start with Schur representation. Consider each eigenvalue in turn. See that there's an invariant subspace (or collection of these) for that eigenvalue. Find Jordan chains and change basis.

The kernel (a subspace) of $(A - \lambda I)^k$ for large k is the generalized eigenspace for λ .

The vectors in this subspaces are called generalized eigenvectors for λ .

Note that ordinary eigenvectors are generalized eigenvectors with depth 1.

$$A \quad \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \quad \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix} \quad \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 7 & 10 \\ 0 & 0 & 3 & 5 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

$$A - 3I \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 7 & 10 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Theorem 9.4.2: Every nilpotent operator N has a basis in which it can be written in block diagonal form with blocks equalling

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & \ddots & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \text{ that is, various sized Jordan blocks with } \lambda = 0$$

Pf. Chose a basis for V consisting of a basis for $\text{Ker}(N)$ and preimages of these vectors.
This is possible because this exhausts the vector space, as N is nilpotent. ■

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Pf. Chose a basis for V consisting of a basis for $\text{Ker}(N)$ and preimages of these vectors. This is possible because this exhausts the vector space, as N is nilpotent. ■

“Everybody gets into the pool.”

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

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$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} =$$

$$\left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right] \begin{pmatrix} a \\ b \end{pmatrix} =$$

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$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a + b \\ b \end{pmatrix}$$

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$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^4 \begin{pmatrix} a \\ b \end{pmatrix} =$$

$$\left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right]^4 \begin{pmatrix} a \\ b \end{pmatrix} =$$

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$$A^4 = (D + N)^4$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

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$$\left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right]^4 \begin{pmatrix} a \\ b \end{pmatrix} =$$

binomial theorem

$$A^4 = (D + N)^4 = D^4 + 4D^3N + 6D^2N^2 + 4DN^3 + N^4$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a+b \\ b \end{pmatrix}$$

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binomial theorem

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binomial theorem

$$A^4 = (D + N)^4 = D^4 + 4D^3N + \cancel{6D^2N^2} + \cancel{4DN^3} + \cancel{N^4} = D^4 + 4D^3N$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a+b \\ b \end{pmatrix}$$

$$\left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right] \begin{pmatrix} a \\ b \end{pmatrix} =$$

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$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^4 \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a+4b \\ b \end{pmatrix} \quad \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right]^4 \begin{pmatrix} a \\ b \end{pmatrix} =$$

binomial theorem

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$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \begin{array}{l} \mathbf{D+N} \text{ (Cor. 9.3.9; Jordan–Chevalley decomposition): Any operator } A \text{ in } V \text{ can be represented as} \\ A = D + N, \text{ where } D \text{ is diagonalizable (i.e. diagonal in some basis) and } N \text{ is nilpotent } (N^m = 0 \text{ for some} \\ m), \text{ and } DN = ND. \end{array}$$

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Corollary to Jordan Canonical Form:

The minimal polynomial of a matrix A divides its characteristic polynomial.

Pf. Consider the JCF of the matrix, which has the same minimal and characteristic polynomials as A .

The minimal polynomial must have factors that include $(\lambda_i - \lambda)$ for each eigenvalue λ_i , and with multiplicity equal to the maximal depth of its generalized eigenvalues. But this polynomial divides the characteristic polynomial. ■

9 things (so far!) that row reduction is great for:

- Solving a system of linear equations
- Checking a set of vectors for: linear independence, spanning, basis
- Computing the inverse of an invertible matrix
- Computing the rank and nullity of a matrix
- Finding factorizations of a matrix: into elementary matrices;
 $A=LU$ and $A=PLU$ and $A=PLDU$ and $A=QR$ factorizations
- Calculating $\det A$
- Finding eigenvalues and eigenvectors of A .
- Finding orthonormal bases.
- Finding orthogonal complements to subspaces.

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- Computing generalized eigenspaces.

$$\det \begin{pmatrix} a & c \\ b & d \end{pmatrix} = ad - bc \qquad \text{tr} \begin{pmatrix} a & c \\ b & d \end{pmatrix} = a + d$$

Facts about the
general 2×2 matrix A

$$\text{If invertible, } \begin{pmatrix} a & c \\ b & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\begin{array}{l} ax + cy = e \\ bx + dy = f \end{array} \Rightarrow \quad x = \frac{\begin{vmatrix} e & c \\ f & d \end{vmatrix}}{\begin{vmatrix} a & c \\ b & d \end{vmatrix}} \quad y = \frac{\begin{vmatrix} a & e \\ b & f \end{vmatrix}}{\begin{vmatrix} a & c \\ b & d \end{vmatrix}}$$

$$\det \begin{pmatrix} a - \lambda & c \\ b & d - \lambda \end{pmatrix} = \lambda^2 - (a + d)\lambda + (ad - bc) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$$

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \Rightarrow \lambda = \frac{\text{tr}(A)}{2} \pm \sqrt{\left(\frac{\text{tr}(A)}{2}\right)^2 - \det(A)}$$

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$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \text{ or } \begin{pmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{pmatrix}$$

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(λ_1, λ_2 not necessarily distinct)

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linear transformations:
 $T(V) \rightarrow W$ such that
 $T(au + bv) = aT(u) + bT(v)$

1

linear transformations are determined
by their action on any basis

2

up to change of bases, every LT between
two vector spaces is determined by its rank

3

linear operators have
a well-defined determinant

4

LO are diagonal in some basis
iff
geom multiplicity = alg multiplicity

5 & 6

every LO is UT in some basis (Schur)
and
every LT has a preferred pair
of orthonormal bases. (SVD)

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every LO has a Jordan canonical form



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matrices / LTs / LOs

$$\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 5 \\ 0 & 0 \end{pmatrix}$$



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rank 0	rank 1	...	rank n
-----------	-----------	-----	-------------



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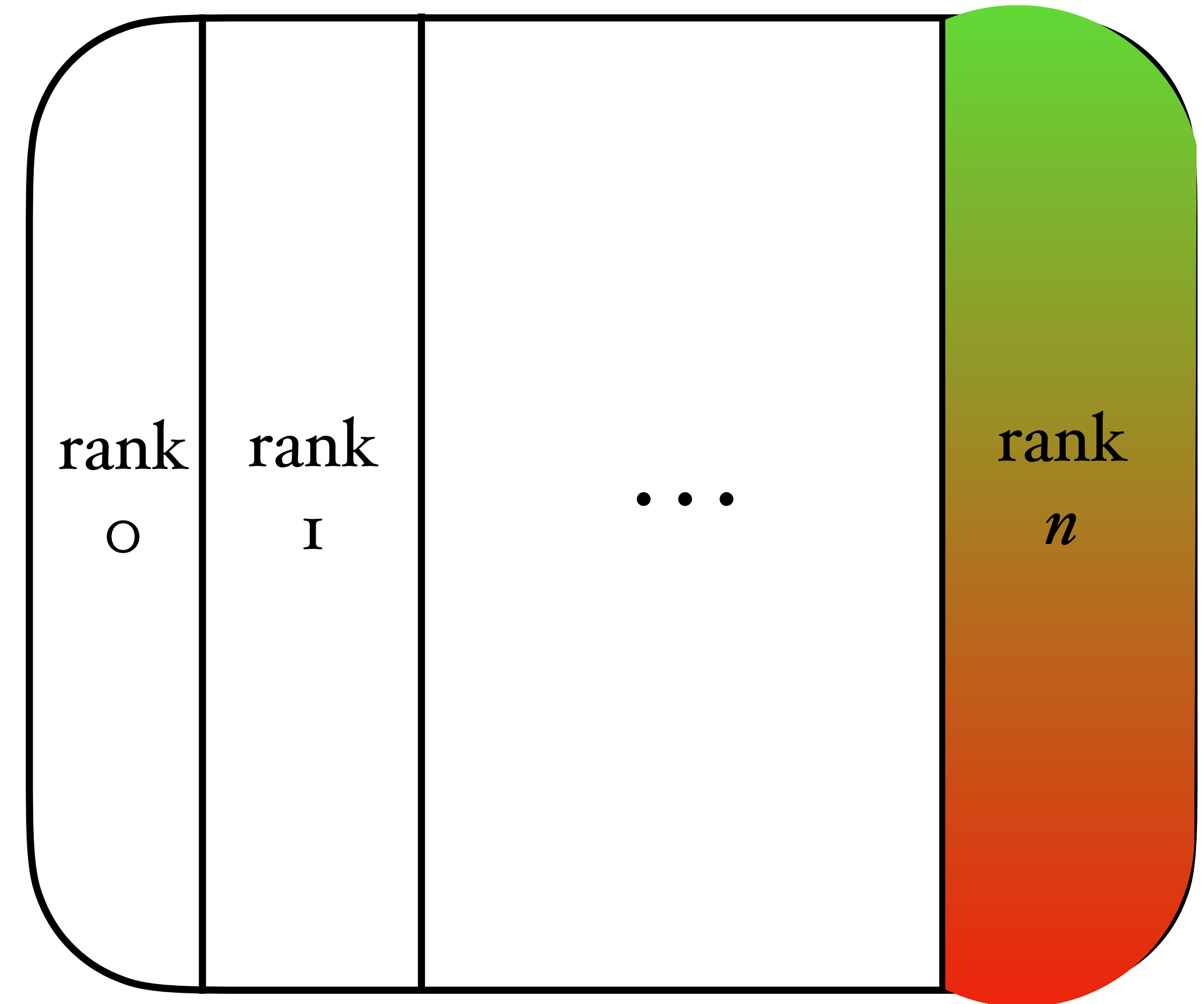
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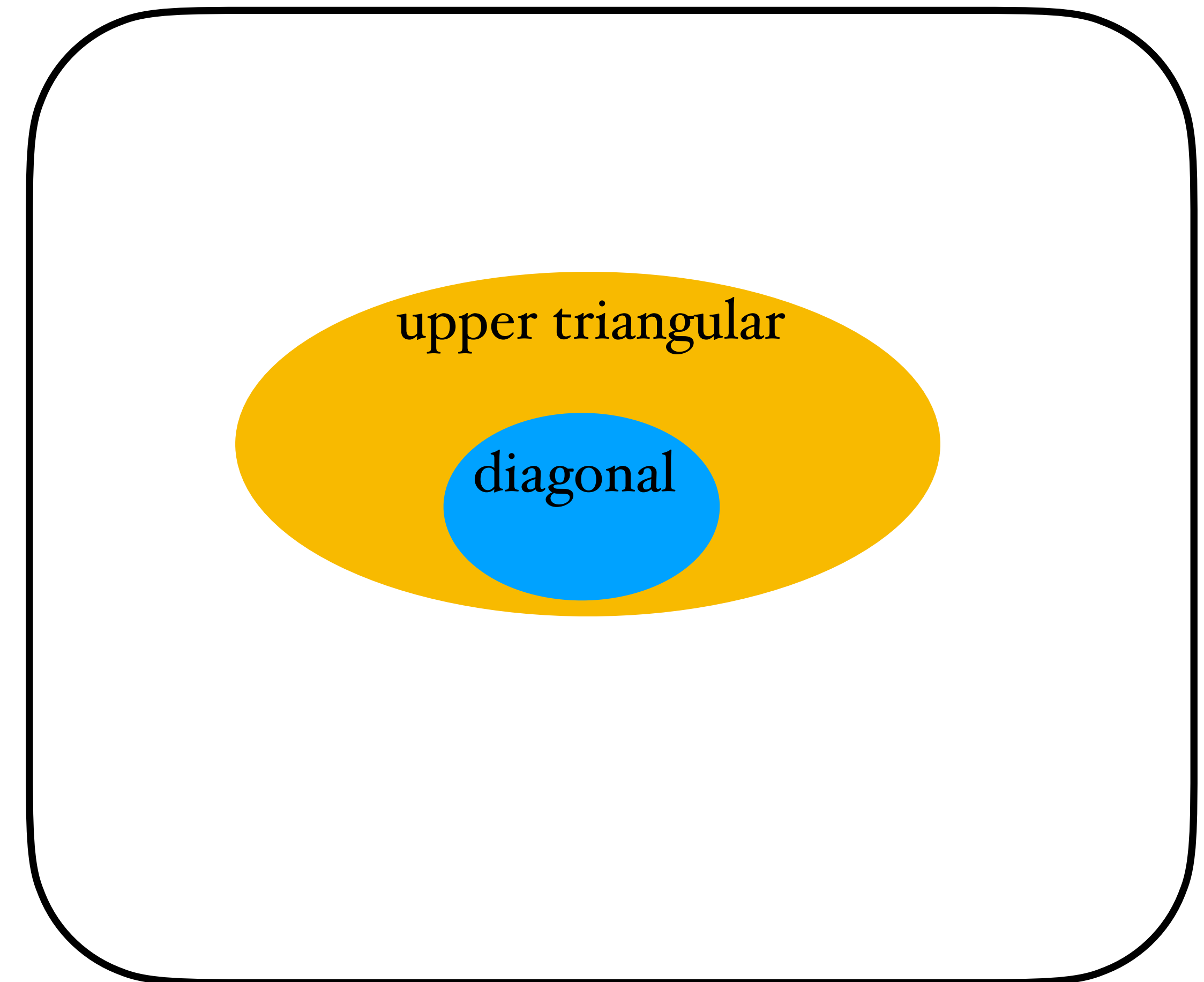
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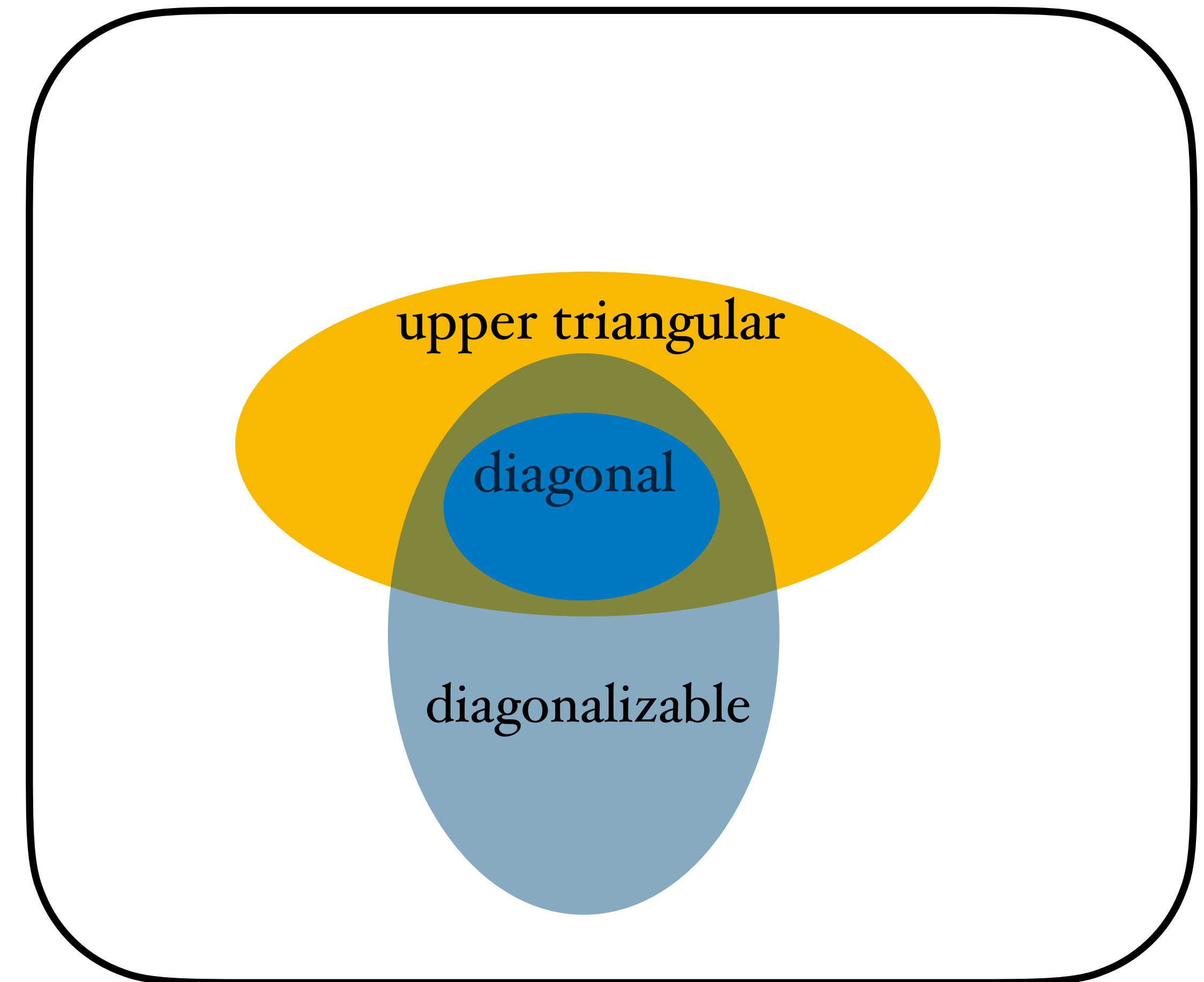
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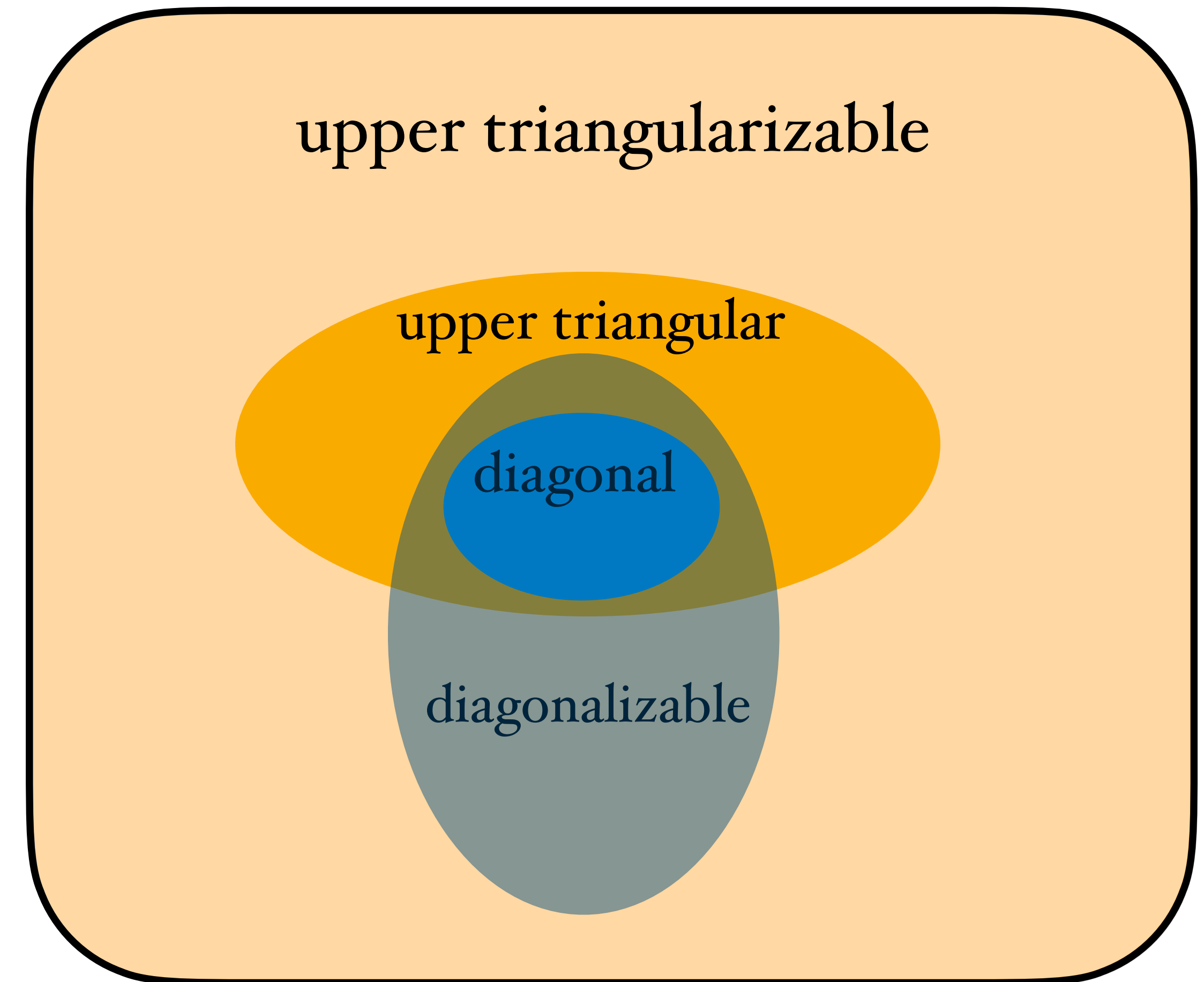
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Jordan Canonical Form

upper triangular

diagonal

diagonalizable

Next up:

- Last requizzing session today, 3-5pm in **Eckhart 206**.
- Flex (or flex plan) due by end-of-day today.
- Final Exam review sessions: Saturday, 3:00-4:30pm, **Kent 120** (packet)
Sunday, 3:00-4:30pm, **Kent 120** (2021 final exam)
Answer keys for each are now in Files under Final.
- End-of-quarter showcase: Monday, 3:00-4:30pm, **Eckhart 202**
Sign up to present by end-of-day today.
- Final Exam: Both exams will be held in our classroom, Ryerson 276.
Section 45 (11:30am section) - Thursday, May 25, 5:30-7:30pm
Section 55 (12:30pm section) - Wednesday, May 24, 7:30-9:30am
- Please fill out the end-of-course survey and the UChicago course evaluation.